AS Level Computer Science — Semester 2 Assessment Pack | Victoria Park Academy

Victoria Park Academy

AS Level Computer Science — Semester 2 Assessment Pack

Academic Year 2024 - 2025

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Total Assessment Marks: 155 (Quizzes 85 + Mock Exam 50 + Practical Portfolio 20)

Assessment Weighting: Quizzes 40% | Mock Exam 35% | Practical Portfolio 25%

Assessment Calendar — Semester 2

Week	Assessment	Topic	Marks	Duration	Format
Week 3	Quiz 1: Algorithms	Searching, Sorting, Algorithm Efficiency	20	25 min	Closed-book
Week 6	Quiz 2: Python Programming	Control Structures, Lists, File I/O	25	30 min	Closed-book (on-screen)
Week 9	Quiz 3: Data Structures	Arrays, Linked Lists, Stacks, Queues	20	25 min	Closed-book
Week 12	Quiz 4: File & Functions	File Handling, Subroutines, Parameter Passing	20	25 min	Closed-book
Week 14	Practical Portfolio Deadline	Programming Project	20	—	Coursework
Week 16	Mock Exam: Paper 2	Problem Solving & Programming	50	60 min	Closed-book

EAL Support Checklist — Assessment Preparation

Key vocabulary list provided 1 week before each assessment

Bilingual (English/Chinese) glossary available during quizzes

Extra reading time: +5 minutes for all timed assessments

Visual diagrams and flowcharts included in all question papers

Clear command words explained: define, explain, compare, evaluate

Sentence starters provided for extended-answer questions

Quiz 1: Algorithms (20 Marks, 25 Minutes)

Topic Coverage: Linear search, binary search, bubble sort, insertion sort, merge sort, algorithm efficiency (time complexity, Big-O notation), pseudocode tracing.

Question 1 — Multiple Choice (4 marks)

Choose the correct answer for each question. Each question is worth 1 mark.

1.1 What is the time complexity of binary search in the worst case?

A) O(n)

B) O(log n)

C) O(n□)

D) O(1)

1.2 Which sorting algorithm has the best average-case time complexity?

A) Bubble sort

B) Insertion sort

C) Merge sort

D) Selection sort

1.3 A linear search is performed on an array of 1000 elements. What is the maximum number of comparisons needed?

A) 10

B) 100

C) 500

D) 1000

1.4 Which of the following is NOT a characteristic of a divide-and-conquer algorithm?

A) Breaks problem into smaller sub-problems

B) Solves sub-problems recursively

C) Always uses iteration rather than recursion

D) Combines solutions to sub-problems

Question 2 — Short Answer (6 marks)

2.1 Explain the difference between linear search and binary search. Include the conditions required for binary search to work. (3 marks)

2.2 Describe how the bubble sort algorithm works. Explain why it is considered inefficient for large datasets. (3 marks)

Question 3 — Algorithm Tracing (6 marks)

Trace the following pseudocode algorithm when called with numbers = [5, 2, 8, 1, 9]:

```
PROCEDURE mystery(numbers: ARRAY OF INTEGER)
FOR i = 0 TO LENGTH(numbers) - 2
FOR j = 0 TO LENGTH(numbers) - 2 - i
IF numbers[j] &gt; numbers[j+1] THEN
temp = numbers[j]
numbers[j] = numbers[j+1]
```

```
numbers[j+1] = temp
ENDIF
ENDFOR
ENDFOR
RETURN numbers
ENDPROCEDURE
```

3.1 Show the state of the array after each complete pass of the outer loop. (4 marks)

3.2 Name this algorithm and state its worst-case time complexity. (2 marks)

Question 4 — Algorithm Comparison (4 marks)

4.1 Compare merge sort and insertion sort in terms of:

Time complexity (best, average, and worst case)

Space complexity

Suitability for nearly-sorted data

(4 marks)

Mark Scheme — Quiz 1: Algorithms

Q1 Multiple Choice (4 marks)

1.1 B — O(log n) □ (1 mark)

1.2 C — Merge sort □ (1 mark)

1.3 D — 1000 □ (1 mark)

1.4 C — Always uses iteration rather than recursion □ (1 mark)

Q2 Short Answer (6 marks)

2.1 Linear search checks each element sequentially (1 mark); binary search repeatedly divides the sorted list in half (1 mark); binary search requires data to be sorted (1 mark)

2.2 Bubble sort repeatedly compares adjacent elements and swaps them if out of order (1 mark); passes through the array until no swaps needed (1 mark); O(n□) worst case makes it slow for large n (1 mark)

Q3 Tracing (6 marks)

3.1 Pass 1: [2, 5, 1, 8, 9] (1 mark); Pass 2: [2, 1, 5, 8, 9] (1 mark); Pass 3: [1, 2, 5, 8, 9] (1 mark); Pass 4: [1, 2, 5, 8, 9] (1 mark)

3.2 Bubble sort (1 mark); O(n□) (1 mark)

Q4 Comparison (4 marks)

Merge sort: O(n log n) all cases, O(n) space, not adaptive (2 marks)

Insertion sort: O(n) best, O(n□) average/worst, O(1) space, adaptive for nearly-sorted (2 marks)

Quiz 2: Python Programming (25 Marks, 30 Minutes)

Topic Coverage: Variables, data types, input/output, selection (if/elif/else), iteration (for, while), list operations, file reading/writing, exception handling.

Question 1 — Code Reading (8 marks)

Study the following Python code:

```
# Program: student_grades.py
def calculate_average(grades):
if len(grades) == 0:
return 0
total = sum(grades)
return total / len(grades)
def get_grade_letter(average):
if average &gt;= 70:
return "A"
elif average &gt;= 60:
return "B"
elif average &gt;= 50:
return "C"
else:
return "F"
# Main program
student_marks = [65, 72, 58, 81, 45]
avg = calculate_average(student_marks)
letter = get_grade_letter(avg)
print(f"Average: {avg:.1f}, Grade: {letter}")
```

1.1 State the data type of student_marks and avg. (2 marks)

1.2 What output will be produced when this program runs? Show your working. (3 marks)

1.3 Explain the purpose of the condition if len(grades) == 0 in the calculate_average function. What error would occur without it? (3 marks)

Question 2 — Code Writing (10 marks)

2.1 Write a Python function called count_vowels that takes a string as a parameter and returns the number of vowels (a, e, i, o, u) in the string. The function should be case-insensitive. (5 marks)

2.2 Write Python code that reads a text file called "data.txt" and prints the number of lines in the file. Include appropriate error handling for the case where the file does not exist. (5 marks)

Question 3 — Debugging (7 marks)

The following program is supposed to find the largest number in a list, but it contains three errors:

```
def find_largest(numbers):
largest = 0
for num in numbers:
if num &gt; largest
largest = num
return num
values = [12, -5, 8, -20, 3]
print(find_largest(values))
```

3.1 Identify and describe the three errors in this code. (6 marks)

3.2 What would be the correct output after fixing all errors? (1 mark)

Mark Scheme — Quiz 2: Python Programming

Q1 Code Reading (8 marks)

1.1 student_marks is a list/array (1 mark); avg is a float/real (1 mark)

1.2 Average = 66.2 (1 mark), Grade = B (1 mark), correct working shown (1 mark)

1.3 Prevents division by zero (1 mark); without it, ZeroDivisionError would occur (1 mark); demonstrates defensive programming (1 mark)

Q2 Code Writing (10 marks)

2.1 Function definition with parameter (1 mark); vowels string defined (1 mark); loop through characters (1 mark); case-insensitive check (1 mark); correct return value (1 mark)

2.2 open() with filename (1 mark); read/count lines (1 mark); print result (1 mark); try-except block (1 mark); FileNotFoundError handling (1 mark)

Q3 Debugging (7 marks)

Error 1: Missing colon after if statement (2 marks)

Error 2: Initialising largest to 0 fails with all negative numbers (2 marks)

Error 3: Returns num instead of largest (2 marks)

Correct output: 12 (1 mark)

Quiz 3: Data Structures (20 Marks, 25 Minutes)

Topic Coverage: Arrays (1D and 2D), linked lists, stacks, queues, records/structs, operations on each structure.

Question 1 — Data Structure Selection (4 marks)

For each scenario, state the most appropriate data structure and justify your choice:

1.1 A web browser needs to implement a "back" button that returns to previously visited pages. (2 marks)

1.2 A printer needs to process documents in the order they were sent. (2 marks)

Question 2 — Stack Operations (6 marks)

A stack is implemented with the following operations: PUSH(item), POP(), and ISEMPTY(). The stack is initially empty.

Perform the following sequence of operations and show the stack contents after each step:

PUSH(5)

PUSH(8)

PUSH(3)

POP()

PUSH(10)

POP()

(6 marks — 1 mark per correct state)

Question 3 — Linked Lists (6 marks)

3.1 Describe two advantages of a linked list over an array. (2 marks)

3.2 Explain why accessing the 5th element in a linked list is slower than accessing the 5th element in an array. (2 marks)

3.3 Draw a diagram showing how a new node is inserted at the beginning of a singly linked list. (2 marks)

Question 4 — 2D Arrays (4 marks)

A 2D array called grid is declared as ARRAY[1:3, 1:4] OF INTEGER.

4.1 How many elements can this array store? (1 mark)

4.2 Write pseudocode to set all elements in the first row to 0. (3 marks)

Mark Scheme — Quiz 3: Data Structures

Q1 Selection (4 marks)

1.1 Stack (1 mark); LIFO matches back button behaviour (1 mark)

1.2 Queue (1 mark); FIFO matches print queue order (1 mark)

Q2 Stack Operations (6 marks)

After PUSH(5): [5] (1 mark)

After PUSH(8): [5, 8] (1 mark)

After PUSH(3): [5, 8, 3] (1 mark)

After POP(): [5, 8] (1 mark)

After PUSH(10): [5, 8, 10] (1 mark)

After POP(): [5, 8] (1 mark)

Q3 Linked Lists (6 marks)

3.1 Any two from: dynamic size, efficient insertion/deletion, no memory waste (2 marks)

3.2 Linked list requires traversal from head (1 mark); array uses direct index access O(1) (1 mark)

3.3 Diagram showing new node pointing to old head, head pointer updated (2 marks)

Q4 2D Arrays (4 marks)

4.1 12 elements (1 mark)

4.2 FOR col = 1 TO 4 (1 mark); grid[1, col] = 0 (1 mark); ENDFOR (1 mark)

Quiz 4: File Handling & Functions (20 Marks, 25 Minutes)

Topic Coverage: Text file operations, CSV processing, functions/procedures, parameter passing (by value/by reference), local/global variables, recursion introduction.

Question 1 — File Operations (6 marks)

1.1 Explain the difference between sequential file access and random (direct) file access. Give one advantage of each. (3 marks)

1.2 A program needs to append new data to an existing file without overwriting the current contents. Describe how this is achieved in Python, including the correct file mode. (3 marks)

Question 2 — Functions and Parameters (8 marks)

2.1 Distinguish between a function and a procedure, giving an example of when each would be used. (2 marks)

2.2 Explain the difference between passing parameters by value and by reference. Include a diagram or example to illustrate your answer. (4 marks)

2.3 Identify whether the following Python code passes the list by value or by reference. Explain your answer.

```
def add_item(items, new_item):

items.append(new_item)

my_list = ["apple", "banana"]

add_item(my_list, "cherry")

print(my_list)
```

(2 marks)

Question 3 — Recursion (6 marks)

The following is a recursive function to calculate factorial:

```
FUNCTION factorial(n: INTEGER) RETURNS INTEGER

IF n

RETURN 1

ELSE

RETURN n * factorial(n - 1)

ENDIF

ENDFUNCTION
```

3.1 Trace the execution of factorial(4), showing each recursive call and the return value at each step. (4 marks)

3.2 State two disadvantages of using recursion compared to iteration. (2 marks)

Mark Scheme — Quiz 4: File Handling & Functions

Q1 File Operations (6 marks)

1.1 Sequential: read/write in order from start (1 mark); random: jump to any location (1 mark); advantage of sequential: simple, efficient for all data (1 mark) OR advantage of random: fast access to specific records

1.2 Open file in append mode 'a' (1 mark); write adds to end without overwriting (1 mark); preserves existing data (1 mark)

Q2 Functions and Parameters (8 marks)

2.1 Function returns a value (1 mark); procedure performs action without returning value (1 mark)

2.2 By value: copy passed, original unchanged (1 mark); by reference: address passed, original modified (1 mark); clear example or diagram (2 marks)

2.3 By reference (1 mark); lists are mutable objects in Python, so modifications affect original (1 mark)

Q3 Recursion (6 marks)

3.1 factorial(4) → 4 * factorial(3) (1 mark); factorial(3) → 3 * factorial(2) (1 mark); factorial(2) → 2 * factorial(1) (1 mark); factorial(1) → 1; returns 24 (1 mark)

3.2 Any two from: higher memory use (stack frames), risk of stack overflow, slower due to function call overhead, harder to debug (2 marks)

End-of-Semester Mock Exam: Paper 2 Style (50 Marks, 60 Minutes)

Examination: AS Level Computer Science — Paper 2 Mock

Time Allowed: 60 minutes

Total Marks: 50

Instructions: Answer ALL questions. Write your answers in the spaces provided. Calculators are not permitted. Show all working for algorithm tracing questions.

Section A: Short Answer (20 marks)

Question 1

1.1 State the output of the following pseudocode when x = 5:

```
IF x MOD 2 = 0 THEN
OUTPUT "Even"
ELSE
OUTPUT "Odd"
ENDIF
```

(1 mark)

1.2 Convert the following pseudocode into Python:

```
DECLARE total : INTEGER
total = 0
FOR i = 1 TO 10
total = total + i
ENDFOR
OUTPUT total
```

(3 marks)

Question 2

2.1 Describe the purpose of a validation check in a program. Give two examples of validation checks. (3 marks)

2.2 Explain the difference between validation and verification. (2 marks)

Question 3

3.1 A program uses a 1D array to store the names of 50 students. State two operations that can be performed on this array. (2 marks)

3.2 Write pseudocode to search for a target name in the array and output its position if found, or "Not found" if not present. (4 marks)

Question 4

4.1 Explain what is meant by "stepwise refinement" in the context of program design. (2 marks)

4.2 Describe the role of a structure diagram in program design. (1 mark)

4.3 Give two benefits of using modules (subroutines) in a large program. (2 marks)

Section B: Problem Solving (18 marks)

Question 5

A teacher wants a program to calculate the final grade for students based on three test scores. The program should:

Input three test scores (each out of 100)

Calculate the average

Assign a grade: A (≥70), B (60 – 69), C (50 – 59), D (40 – 49), F (<40)

Output the average and grade

5.1 Draw a flowchart to represent this algorithm. (6 marks)

5.2 Write the algorithm in pseudocode. (6 marks)

5.3 Describe one additional feature that could be added to validate the input scores. (2 marks)

5.4 The teacher wants to store the results for 30 students. Describe the most appropriate data structure and explain why it is suitable. (4 marks)

Section C: Code Analysis (12 marks)

Question 6

Study the following Python program:

```
def is_prime(n):
if n
return False
for i in range(2, int(n**0.5) + 1):
if n % i == 0:
return False
return True
# Main program
limit = int(input("Enter upper limit: "))
count = 0
for num in range(2, limit + 1):
if is_prime(num):
count = count + 1
print(f"Number of primes: {count}")
```

6.1 State the output when the user enters 10. (2 marks)

6.2 Explain why the loop in is_prime only needs to check up to the square root of n. (3 marks)

6.3 Identify the programming paradigm used in this code and justify your answer. (2 marks)

6.4 Suggest one improvement to make this program more efficient for large values of limit. (2 marks)

6.5 The program currently counts primes. Describe how you would modify it to also output the sum of all primes found. (3 marks)

Mark Scheme — Mock Exam Paper 2

Section A (20 marks)

1.1 "Odd" (1 mark)

1.2 Correct Python syntax: total = 0 (1 mark); for loop 1-10 (1 mark); print(total) (1 mark)

2.1 Ensures data meets criteria (1 mark); range check, type check, presence check, format check (any 2 for 2 marks)

2.2 Validation checks data is reasonable/allowed (1 mark); verification checks data is accurate/correctly entered (1 mark)

3.1 Any two: search, sort, insert, delete, traverse (2 marks)

3.2 Loop through array (1 mark); compare with target (1 mark); output position if found (1 mark); output "Not found" if not (1 mark)

4.1 Breaking problem into smaller sub-problems (1 mark); refining each level until ready to code (1 mark)

4.2 Shows hierarchical structure of modules and their relationships (1 mark)

4.3 Any two: code reuse, easier testing, easier maintenance, team collaboration, abstraction (2 marks)

Section B (18 marks)

5.1 Flowchart with: start/end, input (3 scores), process average, decision (grade boundaries), output (6 marks)

5.2 Pseudocode with: declarations, input, calculation, selection structure, output (6 marks)

5.3 Range check (0-100) or type check (2 marks)

5.4 Array of records/structs (1 mark); stores multiple fields per student (1 mark); allows iteration (1 mark); fixed size known at compile time (1 mark)

Section C (12 marks)

6.1 Number of primes: 4 (2, 3, 5, 7) (2 marks)

6.2 If n has a factor > sqrt(n), the paired factor must be

6.3 Procedural/imperative (1 mark); uses functions and sequential statements (1 mark)

6.4 Sieve of Eratosthenes or store found primes to check divisibility (2 marks)

6.5 Initialise sum variable (1 mark); add prime to sum inside loop (1 mark); output sum after loop (1 mark)

Grade Boundaries

Quiz Grade Boundaries (Raw Marks)

Grade	Quiz 1 (20)	Quiz 2 (25)	Quiz 3 (20)	Quiz 4 (20)
A	16 - 20	20 - 25	16 - 20	16 - 20
B	14 - 15	17 - 19	14 - 15	14 - 15
C	12 - 13	14 - 16	12 - 13	12 - 13
D	10 - 11	11 - 13	10 - 11	10 - 11
E	8 - 9	9 - 10	8 - 9	8 - 9
U	0 - 7	0 - 8	0 - 7	0 - 7

Mock Exam Grade Boundaries (50 marks)

Grade	Marks	UMS Equivalent
A	40 - 50	80 - 100%
B	35 - 39	70 - 79%
C	30 - 34	60 - 69%
D	25 - 29	50 - 59%
E	20 - 24	40 - 49%
U	0 - 19	0 - 39%

Semester 2 Overall Grade Boundaries (155 marks)

Grade	Marks	Percentage
A	124 - 155	80 - 100%
B	109 - 123	70 - 79%
C	93 - 108	60 - 69%
D	78 - 92	50 - 59%
E	62 - 77	40 - 49%

Grade	Marks	Percentage
U	0 - 61	0 - 39%

Practical Portfolio Assessment (20 Marks)

Deadline: Week 14 of Semester 2

Task: Develop a Python program that solves a real-world problem using the concepts covered in Semester 2.

Portfolio Requirements

Component	Marks	Description
Problem Analysis	3	Clear statement of the problem, inputs, outputs, and constraints
Design	4	Structure diagram, pseudocode or flowchart, data structure choices justified
Implementation	6	Working Python code with appropriate use of functions, data structures, and file handling
Testing	4	Test plan with normal, boundary, and erroneous test cases; evidence of testing
Evaluation	3	Reflection on strengths, limitations, and potential improvements

Suggested Project Topics

Bronze: Student mark calculator with file input/output; simple quiz program with score tracking

Silver: Contact management system using lists/dictionaries and file storage; inventory system with search and sort

Gold: Library management system with multiple classes, file persistence, and reporting; simulation using queues and stacks

EAL Portfolio Support

Project brief provided in bilingual format

Sentence starters for analysis and evaluation sections

Visual planning templates for design phase

One-to-one check-in meetings every two weeks

Peer review sessions with structured feedback forms

Assessment Policy and Procedures

Special Considerations

Students with EAL status receive +5 minutes extra time per 30-minute assessment

Bilingual glossaries are permitted in all closed-book quizzes

Students with learning support plans may use assistive technology as specified

All assessment materials are available in large print upon request

Academic Integrity

All submitted work must be the student's own

Plagiarism in the practical portfolio will result in a zero mark and disciplinary action

Collaboration is permitted for peer review only; all code must be individually written

External sources must be cited in the evaluation section

Feedback and Review

Marked quizzes returned within one week with written feedback

Mock exam results discussed in individual target-setting meetings

Portfolio draft feedback provided at Week 10 (formative checkpoint)

Parents receive assessment reports after each quiz cycle

中英双语术语 / Bilingual Assessment Terms:

Algorithm = 算法 | Data Structure = 数据结构 | Function = 函数 | Recursion = 递归 | Validation = 验证 | Pseudocode = 伪代码 | Flowchart = 流程图 | Array = 数组 | Stack = 栈 | Queue = 队列

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